

SHASHANK SINGH

SENIOR PERFORMANCE ENGINEER

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PROFESSIONAL SUMMARY

Senior Performance Engineer with **7.7 years of experience** architecting and executing end-to-end performance testing strategies for enterprise-scale applications across **BFSI, Healthcare, Retail, and Media** domains. Proven expertise in **LoadRunner, JMeter, BlazeMeter, and Cavisson NetStorm** with deep proficiency in APM (**AppDynamics, Dynatrace**), log analytics (**Splunk**), and profiling (**JProfiler, JVisualVM**). Track record of reducing critical production defects by **40%**, improving application response times by **30%**, and driving an up to **94% reduction** in manual workflow execution time through **AI-powered automation**. Experienced in capacity planning, workload modeling, root cause analysis, and embedding performance engineering into **CI/CD pipelines**. Adept at leading teams, mentoring engineers, and presenting data-driven performance insights to senior stakeholders.

KEY ACHIEVEMENTS

- **Built** an Agentic AI workflow that converts LoadRunner scripts to JMeter with 95% accuracy, eliminating weeks of manual migration effort across enterprise engagements.
- **Engineered** an AI-powered Data Preparation Agent that reduced test data setup time from 90 minutes to 5 minutes (a 94% reduction), enabling faster test cycles.
- **Automated** a 14-step manual job processing workflow (data preparation, sequential job triggering, result collection, and Splunk log analysis) reducing execution time from 2 hours to 15 minutes.
- **Built** a natural language Splunk chat assistant that allows teams to query critical performance metrics and stats in plain English, eliminating the need to write SPL queries manually.
- **Achieved** a 40% reduction in critical defects pre-release through systematic performance diagnostics and root cause analysis at Charles Schwab.
- **Optimized** SQL queries and application performance, delivering a 30% improvement in response times for a high-traffic retail platform.

TECHNICAL SKILLS

Performance Testing:	LoadRunner (VuGen, Controller, Analysis), JMeter, BlazeMeter, Cavisson NetStorm, LoadRunner Enterprise (LRE), LoadRunner Cloud
Protocols:	HTTP/HTTPS, SOAP, REST/JSON, Web Services
APM & Monitoring:	AppDynamics, Dynatrace, Perfmon, Splunk (Log Analytics)
Profiling & Debugging:	JProfiler, JVisualVM, Eclipse MAT, Fiddler, Chrome DevTools
CI/CD & Automation:	GitHub Actions, Jenkins, Automated Job Orchestration Pipelines
Programming:	Core Java, MySQL, Groovy
AI & Automation:	Agentic AI Workflows, LLM Integration, RAG, Prompt Engineering, Cursor, GitHub Copilot
Collaboration & Agile:	Rally, Jira, Agile/Scrum, Postman, SoapUI, Rancher

CERTIFICATIONS

- **ISTQB Certified Tester** – Foundation Level
- **Prompt Engineering** – Professional Certification

PROFESSIONAL EXPERIENCE

Senior Performance Engineer | IRIS Software, Noida

Sep 2024 - Present

Client: Charles Schwab (Banking & Financial Services)

- **Spearheaded** the migration of a critical performance testing suite by converting 40 legacy LoadRunner scripts into JMeter and developing 40+ UI workload scenarios within 2 months, reducing tool licensing costs and increasing automation flexibility.
- **Built** an Agentic AI workflow that converts existing LoadRunner scripts to JMeter with 95% accuracy, drastically reducing migration timelines from weeks to hours for future engagements.
- **Engineered** an AI-powered Data Preparation Agent that automates test data setup, cutting preparation time from 90 minutes to 5 minutes per test cycle.
- **Automated** a 14-job sequential processing workflow (previously manual, ~2 hours) into an orchestrated pipeline completing in 15 minutes — including automated result collection, screenshot capture, and Splunk log issue detection post-execution.
- **Architected and deployed** a generative-AI Splunk assistant utilizing natural language processing, allowing cross-functional squads to extract complex production error rates and KPIs without manual SPL query construction.
- **Conducted** deep-dive performance diagnostics leveraging AppDynamics and system monitoring tools to pinpoint bottlenecks, isolate root causes, and resolve memory leaks, achieving a 40% reduction in critical defects pre-release.
- **Designed and executed** SOD/EOD batch job performance validations, analyzing message throughput, processed volumes, response times, and failure rates via Splunk log analytics to drive data-backed optimization decisions.
- **Engineered** robust workload models aligned with production usage patterns, enabling realistic peak-load simulations that directly informed capacity planning and go/no-go production readiness decisions.
- **Embedded** performance engineering into GitHub CI/CD pipelines, enabling automated regression detection on every release and reducing production escalations.

Performance Test Analyst | Coforge Ltd, Noida

Aug 2023 - Sep 2024

Client: Tokio Marine (Insurance - BFSI)

- **Led** performance testing strategy for Agile release cycles, defining non-functional requirements (NFRs), aligning performance acceptance criteria with business SLAs, and delivering accurate effort estimations for sprint planning.
- **Conducted** proof-of-concept evaluations using JMeter to assess performance feasibility for new application modules, influencing architecture decisions early in the development lifecycle.
- **Developed and debugged** automated performance scripts using JMeter and BlazeMeter, covering load, stress, endurance, and scalability scenarios for web-based insurance applications.
- **Performed** deep database performance analysis using Automatic Workload Repository (AWR) reports, identifying slow queries and recommending indexing strategies that improved transaction throughput.

- **Leveraged** AppDynamics for real-time monitoring during load tests and JProfiler for isolated user-level profiling, correlating application-tier bottlenecks with infrastructure constraints.
- **Differentiated** genuine performance scenarios from functional end-to-end flows, optimizing test coverage efficiency and eliminating redundant execution efforts across 3 release trains.
- **Provided** technical mentorship to 2 junior engineers on scripting best practices, correlation techniques, and performance analysis methodologies.

Performance Engineer | Tata Consultancy Services, Bangalore

Dec 2021 - Aug 2023

Client: CVS Health, Rhode Island (Retail/Healthcare)

- **Established** the NFR framework for CVS Health's digital platform, partnering with product owners to translate business KPIs into quantifiable performance benchmarks aligned with SLA commitments.
- **Developed and maintained** performance scripts in VuGen using Web (HTTP/HTML) and Web Services protocols for Java, WebSphere, and web-based application architectures.
- **Designed and executed** comprehensive load, stress, endurance, and scalability tests on LoadRunner Cloud, validating system readiness for seasonal traffic spikes (Open Enrollment, flu season).
- **Performed** application-level profiling under isolated conditions using JProfiler, identifying memory leaks and thread contention issues that improved response times for critical user journeys.
- **Analyzed and interpreted** AWR reports to identify database performance trends, recommending query optimizations and connection pool tuning for Oracle-backed services.
- **Conducted** root cause analysis across application, middleware, and database layers, collaborating with development and infrastructure teams to implement fixes that improved long-term system stability.
- **Validated** API functionality and contract compliance using Postman, ensuring backend service readiness prior to performance test execution.

Performance Tester | Cognizant Technology Solutions, Bangalore/Pune

Sep 2020 - Dec 2021

Client: Family Dollar Stores, Virginia (Retail)

- **Led** cross-functional requirements analysis with business analysts and technical leads, synthesizing system architecture documentation into targeted performance test strategies for a high-traffic retail platform.
- **Designed** workload models and prepared performance test plans during the requirements phase, ensuring test scenarios accurately reflected peak shopping-season traffic patterns.
- **Developed** scripts in Web (HTTP/HTML) and Web Services protocols, designed Controller scenarios, and executed smoke, baseline, and load tests across multiple release cycles.
- **Utilized** AppDynamics APM to monitor server-side metrics (CPU, Memory, Heap, GC) during load tests, correlating client-side response times with backend resource saturation.
- **Collaborated** with development teams to fine-tune SQL queries and application logic, delivering a 30% improvement in response times for order processing workflows.
- **Authored** final Performance Test Summary Reports and presented analysis findings to leadership, influencing go-live decisions for 4 major releases.

Performance Tester | Cognizant Technology Solutions, Bangalore

Oct 2018 - Aug 2020

Client: Warner Bros, Burbank (Media & Entertainment)

- **Built and executed** performance test scripts using HP LoadRunner for Warner Bros' digital content delivery platform, ensuring scalability under global user traffic.
- **Owned** end-to-end test environment readiness — configuring test data, performance tools, and monitoring infrastructure for each test cycle.
- **Validated** API functionality using Postman and executed cloud-based performance tests in LoadRunner Cloud, analyzing results to detect and communicate performance risks to stakeholders.
- **Collected and analyzed** test results including monitoring data, application logs, and server metrics to identify performance bottlenecks and recommend targeted optimizations.

EDUCATION

Bachelor of Technology (B.Tech) in Information Technology • SRM University, Chennai — 2018